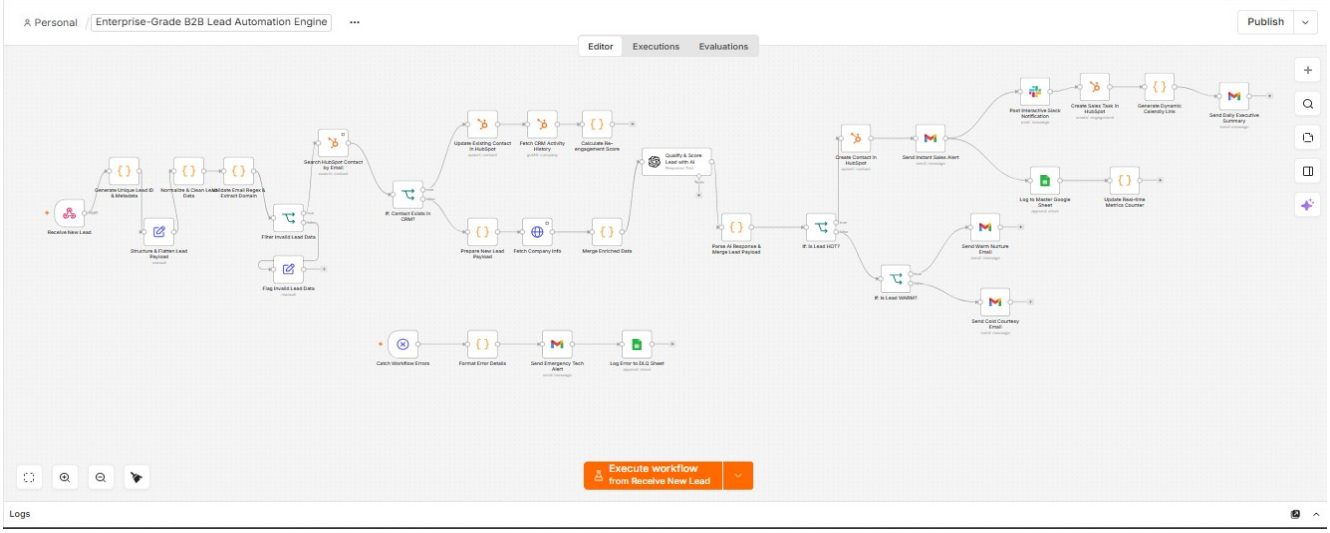
35. PORTFOLIO DOCUMENTATION

AI-Powered B2B Lead Generation, Qualification & HubSpot CRM Automation



Project Overview

A portfolio-grade n8n automation engine for a B2B agency. The workflow receives a lead through a POST webhook, generates a unique lead identifier, structures and cleans the payload, validates email data, checks HubSpot for an existing contact, enriches company information, uses OpenAI for qualification/scoring, routes the lead by tier, performs CRM actions, alerts sales, logs processing information and provides a dedicated error/DLQ path.

Technology Stack

Technology	Role
n8n	Workflow orchestration, transformation, branching and integrations.
HubSpot	Contact search, contact creation/update and sales engagement task path.
OpenAI / GPT-4o-mini	B2B lead qualification, scoring, tiering and recommended action.
Gmail	Sales alerts, nurture/courtesy emails, emergency alerts and executive summary path.
Slack	Interactive HOT-lead notification.
Google Sheets	Master audit log, error/DLQ logging and metrics path.
Clearbit endpoint	Company information enrichment using email domain.
Calendly URL generation	Dynamic discovery-call booking URL creation.

Architecture

Intake: Receive New Lead → Generate Unique Lead ID & Metadata → Structure & Flatten Lead Payload

Data quality: Normalize & Clean Lead Data → Validate Email Regex & Extract Domain → Filter / Flag Invalid Lead Data

CRM deduplication: Search HubSpot Contact by Email → Contact Exists? → Update Existing Contact OR Prepare New Lead Payload

Enrichment + AI: Fetch Company Info → Merge Enriched Data → Qualify & Score Lead with AI → Parse AI Response

Sales routing: HOT/WARM/COLD branches → HubSpot → Gmail/Slack → sales task → Calendly link where applicable.

Operations: Master Google Sheet audit log → metrics path → executive summary path.

Reliability: Catch Workflow Errors → Format Error Details → Emergency Tech Alert → Log Error to DLQ Sheet.

33 Configured Nodes

1. Receive New Lead	18. If: Is Lead WARM?
2. Generate Unique Lead ID & Metadata	19. Send Warm Nurture Email
3. Structure & Flatten Lead Payload	20. Send Cold Courtesy Email
4. Normalize & Clean Lead Data	21. Update Existing Contact in HubSpot
5. Validate Email Regex & Extract Domain	22. Log to Master Google Sheet
6. Filter Invalid Lead Data	23. Catch Workflow Errors
7. Flag Invalid Lead Data	24. Format Error Details
8. Search HubSpot Contact by Email	25. Send Emergency Tech Alert
9. If: Contact Exists in CRM?	26. Log Error to DLQ Sheet
10. Prepare New Lead Payload	27. Post Interactive Slack Notification
11. Fetch Company Info	28. Create Sales Task in HubSpot
12. Merge Enriched Data	29. Generate Dynamic Calendly Link
13. Qualify & Score Lead with AI	30. Fetch CRM Activity History
14. Parse AI Response & Merge Lead Payload	31. Calculate Re-engagement Score
15. If: Is Lead HOT?	32. Update Real-time Metrics Counter
16. Create Contact in HubSpot	33. Send Daily Executive Summary
17. Send Instant Sales Alert	

AI Qualification Contract

The AI qualification node receives business-relevant lead attributes including full name, job title, company, company size, budget, timeline, service needed and notes. The supplied workflow expects a raw JSON response.

Field	Purpose
lead_score	Numerical qualification score.
lead_tier	Routing class such as HOT/WARM.
qualification_reason	Human-readable explanation of the AI decision.
recommended_action	Next action for the sales team.

Reliability & Recovery

The workflow includes a dedicated n8n Error Trigger. Error details are transformed into failed node, error message, execution ID, execution URL and timestamp fields, followed by an emergency technical alert and a DLQ-style Google Sheet log. This gives the operator a traceable failure record.

Portfolio Positioning

This project demonstrates more than basic n8n connectivity. It combines event-driven intake, data quality, CRM deduplication, company enrichment, AI decisioning, sales routing, multi-channel notification, audit logging and failure observability.

Recommended Case-Study Headline

Built an AI-powered B2B Lead Operations Engine that validates, enriches, qualifies and routes inbound leads into HubSpot—while keeping sales teams informed and failures traceable.

Technical Skills Demonstrated

n8n workflow architecture • Webhooks • JavaScript/Code nodes • Data normalization • Regex validation • Conditional routing • HubSpot CRM operations • OpenAI integration • HTTP APIs • Gmail automation • Slack notifications • Google Sheets logging • Error Trigger • DLQ-style failure logging • Dynamic URL generation • Operational metrics.

Implementation Note

This document reflects the supplied workflow JSON and screenshot. The current build contains 33 configured nodes. Some auxiliary paths visible in the workflow—such as re-engagement scoring, metrics and executive-summary logic—can be further connected and hardened for a production deployment. The project is best presented as a portfolio automation build with a clear path to production hardening.